Proper Hat Guessing on Friendship Graphs: The Hourglass Value and the Three-Petal Threshold

Ryutaro Yonezu (Yonezu Research Lab)

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1. Model and exact-value candidate

Let

$$F_r = K_1 \vee (rK_2), \qquad r \geq 1.$$

The hub is v, and the leaves of petal i are u_i, w_i. Its edges are vu_i, vw_i, u_iw_i, for $1 \leq i \leq r$, with no others. Thus F_r consists of r triangles with one common vertex. It is also the triangular windmill $W_{3,r}$; F_2 is the hourglass or bowtie graph. These names do not denote two triangles sharing an edge.

The palette is $Q = \{0, \dots, q-1\}$. An adversary chooses a proper vertex coloring: adjacent vertices have distinct colors. Each player sees its own open neighborhood, makes one simultaneous deterministic guess for its own color, and the team wins if at least one guess is right. Different vertices may have different functions and may distinguish their neighbors. No anonymous, affine, cyclic, or symmetry restriction is imposed on the model.

Write $\mathrm{HG}_P(G)$ for the largest winning palette size.

Main result candidate. The hourglass has $\mathrm{HG}_P(F_2) = 6$, and three is the smallest positive number of petals for which seven colors are winning. Together with the known triangle value, the uniform upper bound, and the extension argument below, this gives, for every integer $r \geq 1$,

$$\mathrm{HG}_P(F_r) = \begin{cases} 5, & r = 1, \\ 6, & r = 2, \\ 7, & r \geq 3. \end{cases}$$

The first case is already the complete-graph theorem of Adriaensen et al. [1]. In [1, v4, Table 1], the hourglass is listed with bounds 5 and 7. The possible contribution of this note is its specific evaluation at $r = 2$ and the sharp three-petal saturation statement for this established model. It does not claim invention of the game, of friendship graphs, or of a general observation-completion method.

2. Local functions and a pair-cover criterion

Write $L_i(a, y)$ for u_i's guess when the hub has color a and w_i has color y, and $R_i(a, x)$ for w_i's corresponding rule. For $q \geq 3$, we may assume

$$L_i(a, y) \notin \{a, y\}, \qquad R_i(a, x) \notin \{a, x\} \quad (a \neq y,\ a \neq x).$$

An output equal to a visible color is never correct under a proper coloring, so replacing it by any legal color cannot destroy a win. Every distinct pair (a, y) is an attainable leaf view in F_r.

For fixed leaf rules, define a graph E_i on the palette Q. Its edge $\{a, b\}$, with $a \neq b$, is present exactly when, for every ordered pair of distinct colors $x, y \in Q \setminus \{a, b\}$,

$$L_i(a, y) = x \ \vee \ R_i(a, x) = y \ \vee \ L_i(b, y) = x \ \vee \ R_i(b, x) = y. \tag{1}$$

Lemma 1 (pair-cover criterion). The fixed leaf rules admit a winning hub rule if and only if $E_1 \cup \cdots \cup E_r = K_q$.

Proof. If a pair $\{a, b\}$ is absent from every E_i, choose in each petal distinct colors $x_i, y_i \notin \{a, b\}$ for which all four equalities in (1) fail. With this common observed leaf vector, both hub colors a, b give proper colorings and make all leaves wrong. The hub sees the same vector in both colorings, so it must be wrong in at least one. Thus its rule cannot be winning.

Conversely, for an observed leaf vector $p = ((x_i, y_i))_{i=1}^r$, put

$$S(p) = \{a \in Q : \ a \notin \{x_i, y_i\}, \ L_i(a, y_i) \neq x_i, \ R_i(a, x_i) \neq y_i \text{ for every } i\}.$$

If two different colors a, b belonged to $S(p)$, the pair would be absent from every E_i, contrary to the assumed cover. Hence $|S(p)| \leq 1$. The hub guesses its sole element when one exists, and guesses arbitrarily otherwise. Whenever all leaves are wrong, the actual hub color belongs to $S(p)$, so the hub is right. Its rule depends only on the leaf colors it sees, not on its own color. This is a simultaneous local strategy and requires no communication during play. □

This is a specialization of the familiar idea of completing the rule at a common vertex after fixing rules on independent pieces; related frameworks appear in [2, Lemma 4.4] and [3, Lemma 3.3]. No claim that this general idea is new is intended.

3. Eight colors are impossible for every number of petals

Lemma 2. For all $r \geq 1$, $\mathrm{HG}_P(F_r) \leq 7$.

Proof. At $q = 8$, fix any two distinct prospective hub colors a, b. There are $6 \cdot 5 = 30$ ordered distinct leaf-color pairs in $T = Q \setminus \{a, b\}$. For one petal, each of the four equalities in (1) holds for at most six such pairs: for example, fixing y determines at most one $x = L_i(a, y)$. The four events therefore cover at most $4 \cdot 6 = 24 < 30$ pairs. Thus $\{a, b\}$ is absent from every E_i. Lemma 1 excludes a winning strategy, regardless of r.

A winning strategy on a larger palette would restrict to one on eight colors: retain guesses inside the restricted palette and replace other outputs arbitrarily. Correct guesses on restricted colorings are preserved. Therefore no larger palette is winning either. □

4. Seven-color rigidity excludes the hourglass

The next argument treats arbitrary legal functions L, R for a single petal. In particular it does not assume the symmetric cyclic form of the constructions below.

Lemma 3. At $q = 7$, the pair graph E of one petal has maximum degree at most 3, and hence at most 10 edges.

Proof. Suppose $\{a, b\} \in E$, and let $T = Q \setminus \{a, b\}$. There are 20 ordered pairs of distinct elements of T. The four correctness events in (1) each have size at most five. Since their union covers all 20 pairs, each event has size exactly five, and the four events are pairwise disjoint.

In particular, $L(a, y)$ belongs to $T \setminus \{y\}$ for every $y \in T$: otherwise the event $L(a, y) = x$ would have fewer than five pairs. Together with legality at $y = b$, this shows that $L(a, y) \neq b$ for every $y \neq a$. Likewise $R(a, x) \neq b$ for every $x \neq a$. Thus each neighbor b of a in E is omitted from the entire images of both functions at hub color a.

If a had four distinct neighbors forming a set B, all outputs of $L(a, \cdot)$ and $R(a, \cdot)$ would lie in the two-element set $U = Q \setminus (\{a\} \cup B) = \{s, t\}$. Legality forces $L(a, t) = s$ and $R(a, s) = t$. Choose any $b \in B$. Then $(x, y) = (s, t)$ is one of the 20 pairs for $\{a, b\}$, and both events $L(a, y) = x$ and $R(a, x) = y$ hold. This contradicts their disjointness. Consequently the maximum degree is at most 3, giving $|E| \leq \lfloor 7 \cdot 3/2 \rfloor = 10$. □

Corollary 4. $\mathrm{HG}_P(F_2) \leq 6$.

Proof. Two seven-color petal graphs cover at most 20 edges in total, whereas K_7 has 21 edges. Apply Lemma 1, then palette restriction to exclude all larger palettes as well. □

The equality condition in Lemma 3 is essential. Merely bounding each event by five, without deriving disjointness, does not prove this result. The number 10 is an upper bound on the number of edges, not an asserted attainable optimum. No construction attaining 10 edges, or optimization certificate for that objective, is needed for Corollary 4.

5. An explicit six-color strategy on two petals

Use arithmetic modulo 6. Define functions on nonzero differences by

d	1	2	3	4	5
$A(d)$	4	3	1	3	3
$B(d)$	4	5	4	5	1

Both leaves in the first petal guess $f_A(a, y) = a + A(y - a) \pmod 6$, and both leaves in the second use $f_B(a, y) = a + B(y - a) \pmod 6$. All outputs are legal.

The A-petal covers every hub pair whose difference is ± 1 or ± 2; the B-petal covers every pair of difference 3. Translation invariance reduces verification to hub pairs $\{0, 1\}$, $\{0, 2\}$, and $\{0, 3\}$, respectively.

For a symmetric leaf rule, an unordered leaf-color pair $\{x, y\}$ is covered at hub color a if $f(a, y) = x$ or $f(a, x) = y$. The following lists of covered unordered pairs verify all three cases directly. A pair such as 23 denotes $\{2, 3\}$, not the integer twenty-three.

Hub pair; rule	Remaining leaf palette	Covered at hub 0	Covered at the other hub color
$\{0, 1\}; A$	$\{2, 3, 4, 5\}$	23, 34, 35	25, 34, 24, 45
$\{0, 2\}; A$	$\{1, 3, 4, 5\}$	14, 13, 34, 35	15, 45, 35
$\{0, 3\}; B$	$\{1, 2, 4, 5\}$	14, 25, 45, 15	12, 24, 14, 25

In each row the union is all six pairs of the remaining four colors. The pair graphs cover K_6, so Lemma 1 supplies the hub rule. This proves the lower bound 6, completing the exact hourglass value with Corollary 4.

6. An explicit seven-color strategy on three petals

Use arithmetic modulo 7, and set

t	1	2	3	4	5	6
$h(t)$	3	5	2	5	3	2

Index the three petals by $d \in \{1, 2, 3\}$. Both leaves of petal d use

$$f_d(a, y) = a + d\, h(d^{-1}(y - a)) \pmod 7. \tag{2}$$

Here d^{-1} is the inverse modulo 7; the argument of h is never zero in a proper local view. The values of h differ from both zero and their arguments, so (2) is legal.

We show that petal d covers all hub pairs of difference $\pm d$. Translation and scaling reduce this to $d = 1$, hub pair $\{0, 1\}$. The remaining colors are $T = \{2, 3, 4, 5, 6\}$. For $y \in T$, the two guesses at hypothetical hub colors 0 and 1 are:

y	$f_1(0,y)$	$f_1(1,y)$
2	5	4
3	2	6
4	5	3
5	3	6
6	2	4

The ten directed pairs from a row label to its two outputs contain exactly one orientation of each unordered pair of distinct elements of T. Thus, for every ordered distinct $(x, y) \in T^2$, either $x \in \{f_1(0, y), f_1(1, y)\}$, or $y \in \{f_1(0, x), f_1(1, x)\}$. This is precisely (1).

The three differences $\pm1, \pm2, \pm3$ exhaust the six nonzero differences modulo 7, so the three petal graphs cover K_7. Lemma 1 gives the hub rule, proving a winning seven-color strategy. For $r > 3$, retain this strategy on three selected petals: the hub ignores its additional neighbors and all added leaves may guess arbitrarily. Every proper coloring restricts to a proper coloring on the selected F_3. Lemma 2 now proves the exact value 7 for every $r \geq 3$.

7. The one-petal case and computational evidence

Since $F_1 = K_3$, [1, Theorem 1.1] gives the already known value 5. It also follows from a simple upper count and a finite construction. At $q = 6$, there are $6 \cdot 5 \cdot 4 = 120$ proper triangle colorings; each player can be right in at most $6 \cdot 5 = 30$ of them, so three players cannot cover all colorings. A five-color construction is included in `witness.json` for reproducibility.

The finite lower-bound checks enumerate, respectively,

$$5(4 \cdot 3) = 60, \quad 6(5 \cdot 4)^2 = 2400, \quad 7(6 \cdot 5)^3 = 189000$$

proper colorings. `verify_exhaustive.py` constructs an actual hub lookup table before checking all proper colorings. `audit_bitmasks.py` separately checks intersections of per-petal surviving-hub bit masks over all outer views. Neither requires a SAT solver, network access, paid API, or extra Python packages. They verify explicit constructions, not a search over all possible strategies. The universal upper bounds are the mathematical arguments in Sections 3–4. Finite computation is unnecessary for their universal quantifiers.

Exploratory SAT searches helped find the tables. Several unrestricted searches ended UNKNOWN and provide no nonexistence evidence. Cyclic and leaf-symmetric restrictions were used only to simplify lower-bound constructions. The delivered proofs do not rely on SAT status, floating point calculations, or timeouts.

8. Prior-art and claim limits

The proper-coloring model, complete-graph value, and the initial hourglass bounds are due to [1]. Ordinary windmill hat guessing is treated in [2], and a proper-coloring coverability framework for a different family of book graphs is developed in [3]. A recent general upper bound is [4]. These are essential comparisons, not evidence of a newly invented model.

The portions of references [1]–[4] cited here were checked in their primary texts by the originating assistant on 23 October 2026. The exact locations and the distinction between ordinary windmills and proper books are recorded in `REFERENCE_CHECK_v0_2.md`. This is a check of the relevant passages, not a line-by-line audit of all proofs in all four papers.

Eventual attainment of seven by itself is a weaker contribution than the sharp threshold. The multiple-guess remark in [1, Section 7, following Problem 7.1] supplies a five-color proper strategy on an edge with two guesses at each endpoint. Alternatively, this fact follows directly by letting both endpoints guess the observed color plus 1 and plus 2 modulo 5. Assign one petal to each of the 21 unordered pairs of prospective hub colors and use the first or second guess according to that hub color. On the other five colors this petal excludes simultaneous failure for both hub candidates. Lemma 1 then gives a seven-color strategy on 21 petals. This is our elementary deduction from a known result, not a theorem about friendship graphs claimed to appear in [1]. Details are in `REFERENCE_CHECK_v0_2.md`. The claimed contribution remains the exact hourglass value and the least-petal threshold three.

Targeted searches through 22 October 2026 did not locate this specific evaluation. That is a bounded search result, not proof of worldwide novelty. The user-supplied external-AI report and its evidence limits are preserved in `audit/USER_SUPPLIED_AUDIT_20261023.md` and `AUDIT_RESPONSE_v0_2.md`. It does not supply the reviewer's CP-SAT program or raw solver logs, and we do not claim to have rerun those computations. No priority, human refereeing, or proof-assistant verification claim is made. The author authorized public preprint release on 23 October 2026. The frozen audit archive retains its original historical HOLD statements. See `PRIOR_ART.md`, `READ_FIRST.md`, and the accompanying publication-status addendum for the distinction between that history and the present release.

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Research inquiries: yonezaemon1+research@gmail.com

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